Ann Von Holle

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Population Health Scientist

Doctoral-level epidemiologist and biostatistician with 15+ years designing, analyzing, and publishing observational and real-world evidence (RWE) research across academia, government, and healthcare payer/life-sciences settings. Lead author of multiple peer-reviewed publications spanning epidemiologic methods, biostatistics, predictive modeling, and machine learning. Proven track record translating large-scale claims, cohort, and genomic data into published research supporting population health outcomes and clinical decision-making.

Core competencies

epidemiologic methods, biostatistics, real-world evidence (RWE), scientific writing, predictive modeling, unsupervised machine learning, clustering, observational studies, data visualization, R programming

Research Experience

American Medical Group Association, Remote **February 2025 – Present**

Senior Population Health Research Analyst

Scope: Lead real-world evidence (RWE) research within the AMGA Research and Analytics group, owning the full research lifecycle from study ideation through peer-reviewed publication and dissemination.

- Develop novel study concepts and research protocols addressing population health priorities, translating findings into peer-reviewed manuscripts and conference presentations.
- Design and conduct analyses using real-world data and advanced epidemiological and analytic methods as part of a collaborative process.
- Apply advanced analytic skills to design statistically sound analytical approaches and compelling data visualizations supporting population health improvement for medical groups and integrated health care systems across the US.
- Collaborate with multiple stakeholders including sponsors, subject matter experts, members of the programming team, and other analysts to ensure projects are completed successfully.

Optum Genomics, Remote **April 2023 - July 2024**

Associate Research Director

Scope: Conducted real world evidence (RWE) studies and decision analytic modeling to evaluate health and economic outcomes, focusing on genetic diagnostic test validity and cancer detection. Contributed to white papers and literature reviews on molecular diagnostics.

- Planned, designed, and executed a simulation study on carrier screening to evaluate health and economic outcomes, leveraging decision analytic models and R software.
- Conducted decision analytic modeling using R software, incorporating clinical validity measures, simulations, and claims data to support evidence-based conclusions for carrier screening.
- Contributed to research focusing on cancer early detection, evaluating the analytical and clinical validity of molecular diagnostic tests by analyzing medical claims data with R software.
- Collaborated on the development of white papers addressing the analytical and clinical validity of genetic and molecular lab-developed tests, providing expert insights and recommendations.
- Conducted comprehensive literature reviews on the quality of molecular diagnostic tests, integrating findings to inform research and practice.

National Institute of Environmental Health Sciences, Research Triangle Park, NC
Special Volunteer **April 2023 – April 2026**
Complete manuscript submission process for research completed during my postdoctoral fellowship.

Postdoctoral Fellow **July 2018 - April 2023**
Biostatistics and Computational Biology Branch, Mentor: Clarice R. Weinberg

Scope: Conducted observational research studies by developing research concepts, specifying statistical analysis plans, doing formal analyses, and authoring manuscripts related to breast cancer outcomes and serum iron biomarkers. Used advanced analytical techniques and high-performance cluster computing, communicated findings at national conferences, and mentored students.

- Developed research concepts and hypotheses, planned eight lead-author manuscripts, and selected appropriate data and analytic methods as part of a collaborative process.
- Applied advanced analytic approaches such as time-to-event analysis, longitudinal mixed-effects models, machine learning, meta-analyses, and data visualization in research on breast cancer outcomes and biomarker exposures.
- Authored initial drafts and reviewed subsequent drafts for epidemiological research manuscripts
- Presented research findings internally and at national conferences.
- Performed large-scale data analyses using a high-performance computing environment
- Mentored graduate and undergraduate students in developing analysis plans, conducting research, and presenting results as part of NIH training programs.

University of North Carolina, Chapel Hill, Chapel Hill, NC
Graduate Researcher **August 2013 - June 2018**
Department of Epidemiology, Advisor: Kari E. North

Scope: As a doctoral candidate in a genetic epidemiology training program, used latent growth mixed effects modeling to assess infant anthropometric growth and its association with adolescent lipid levels in relation to sociodemographic factors and genetic risk variants. Authored three lead manuscripts, secured a predoctoral grant from the American Heart Association, and used reproducible research techniques with R, Markdown, LaTeX, and Git.

- Using mixed effects models and latent growth modeling, characterized infant anthropometric growth as it relates to sociodemographic risk factors, its impact on lipid levels in adolescence, and its role as an effect modifier in the association between genetic risk variants and lipid levels. Developed scientific hypotheses, created analysis plans, analyzed observational cohort data, and wrote manuscripts resulting in three lead author manuscripts.
- Employed advanced statistical techniques such as latent growth mixed effects modeling, factor analysis, and non-linear random effects models to estimate patterns of infant growth and their association with adolescent lipid levels
- Developed and submitted a pre-doctoral American Heart Association grant application titled “Infant Growth Trajectories and Dyslipidemia in Adolescence,” funded in 2016 (Award Number 16PRE29200008).
- Implemented reproducible research techniques by documenting writing and statistical analyses in R, Markdown, and LaTeX, and maintained version control with Git.

Biostatistician, Chapel Hill, NC **September 2005 - September 2014**
Department of Psychiatry, Center of Excellence for Eating Disorders (CEED)

Scope: Worked across various academic levels and collaborated with researchers to plan, conduct, and interpret statistical analyses for peer-reviewed manuscripts

- Collaborated in the planning and interpretation of statistical analyses for research by faculty, post-doctoral fellows, and students.
- Analyzed observational, clinical trial, and medical record data using regression and structural equation models in SAS and R, contributing to over forty peer-reviewed publications.
- Authored statistical methods sections, created data visualizations, and ensured data integrity for peer-reviewed publications.

Education

Doctor of Philosophy (Ph.D.), Epidemiology, University of North Carolina at Chapel Hill, Chapel Hill, NC | **Thesis:** Infant Growth Trajectories and Lipid Levels in Adolescence
Master of Science (M.S.), Biostatistics, University of North Carolina at Chapel Hill, Chapel Hill, NC
Bachelor of Arts (B.A.), Applied Mathematics, University of California, San Diego, San Diego, CA

Recent Select Publications

*Indicates a mentored student's work. †Indicates equal authorship

1. **Inflammatory marker profiles vary by BMI across Crohn's disease, ulcerative colitis, psoriasis, and psoriatic arthritis**.
   Shields, S. A., **Von Holle, A.**, Padival, R., Merola, J. F., Zuckerman, R., Mohl, J. T., & Ciemins, E. L. (2026) *Frontiers in Endocrinology*, *17*, 1851905. https://doi.org/10.3389/fendo.2026.1851905
2. **Primary adherence to medications for the treatment of major depressive disorder: A descriptive study using real-world data.**
   Von Holle, A., Mohl, J. T., Zuckerman, R. B., Obenrader, J., & Ciemins, E. L. (2026). Journal of Managed Care & Specialty Pharmacy, 32(7), 793–806.
   https://doi.org/10.18553/jmcp.2026.32.7.793
3. **Sedentary Time and Breast Cancer Risk in the Sister Study Cohort.**
   Donzella, Sidney M., Rogers M, O'Brien KM, **Von Holle A**, Sandler DP, Weinberg CR, and Díaz Santana MV. (2025); Cancer Epidemiology, Biomarkers & Prevention 34(11):1955–62. doi:10.1158/1055-9965.EPI-25-0510
4. **Association Between Body Iron Status and Biological Aging.**
   Von Holle A, Ramamurthy S*, Díaz Santana MV, et al. *Nutrients*. (2025);17(9):1409. doi:10.3390/nu17091409
5. **BMI and Breast Cancer Risk around Age at Menopause**
   Von Holle A, Adami HO, Baglietto L, et al. Cancer Epidemiology. 89 (2024):102545, DOI: 10.1016/j.canep.2024.102545
6. **Infant Growth Trajectories and Lipid Levels in Adolescence: Evidence From a Chilean Infancy Cohort**
   Von Holle A, North KE, Gahagan S, Blanco E, Burrows R, Lozoff B, Howard AG, Justice AE, Graff M, and Voruganti S. American Journal of Epidemiology. (2022) :kwac057, DOI: 10.1093/aje/kwac057
7. **Does This Adult Patient Have Hypertension?: The Rational Clinical Examination Systematic Review.**
   Viera AJ, Yano Y, Lin FC, Simel DL, Yun J, Dave G, **Von Holle A**, Viera LA, Shimbo D, Hardy ST, Donahue KE, Hinderliter A, Voisin CE, and Jonas DE.
   JAMA. 326 (2021) :339–47, DOI: 10.1001/jama.2021.4533
8. **Evidence for familial clustering in breast cancer age of onset.**
   Von Holle A, O'Brien KM, Sandler DP, and Weinberg CR.
   International Journal of Epidemiology. 50 (2021) :97–104, DOI: 10.1093/ije/dyaa201

Full publication list available at
https://www.ncbi.nlm.nih.gov/myncbi/ann.von%20holle.1/bibliography/public/

Publications In Preparation

1. Associations between male-origin microchimerism and breast, ovarian, and uterine cancer. (2026)
   Von Holle, Ann, O'Brien KM, Sandler DP, Weinberg CR

Grants and awards

Grants
Individual Pre-Doctoral Fellowship Award (Award: 16PRE29200008)
American Heart Association (Total: $51,900) 2016 - 2018

Awards
Intramural Paper of the Month
Environmental Factor, NIEHS Newsletter May 2021
Title: Evidence for familial clustering in breast cancer age of onset.

Oral presentations

*Represents trainee's work for which Ann Von Holle was a mentor.

Association between Body Iron Status and Biological Aging, Durham, NC 2023
Ramamurthy S*, Von Holle A, Diáz-Santana, Kresovich J, Taylor J, Xu Z, O'brien Km, Sandler Dp, Weinberg Cr
National Institute of Environmental Health Sciences Scholars Connect Spring Symposium.

Infant Growth Trajectories and Lipid Levels in Adolescence: Evidence from a Chilean Infancy Cohort, New Orleans, LA
Von Holle A, North Ke, Gahagan S, Blanco, E, ET AL. American Heart Association EPI/Lifestyle Sessions 2018

Poster presentations

*Represents trainee's work for which Ann Von Holle was a mentor.

Associations between Serum Iron Biomarkers and Breast Cancer Tumor Size, Orlando, FL 2023
VON HOLLE A, THOMPSON RL*, O'BRIEN KM, SANDLER DP, WEINBERG CR American Association of Cancer Research Annual Meeting

Association between Body Iron Status and the Aging Process, Virtual 2022
RAMAMURTHY S*, **VON HOLLE A**, DIÁZ-SANTANA, KRESOVICH J, TAYLOR J, XU Z, O'BRIEN KM, SANDLER DP, WEINBERG CR
National Institute of Environmental Health Sciences Scholars Connect Poster Sessions

Characterizing the association between BMI and breast cancer risk by menopause status, Chicago, IL 2022
And age-time **VON HOLLE A**, O'BRIEN KM, SANDLER DP, SCHOEMAKER M, SWERDLOW A, NICHOLS HB, JONES ME, JANICEK R, THE PREMENOPAUSAL BREAST CANCER COLLABORATIVE GROUP, WEINBERG CR
Society for Epidemiologic Research

Teaching and Mentoring Experience

National Institute of Environmental Health Sciences, Research Triangle Park, NC
- Mentor, 1 undergraduate student, NIEHS Scholars Connect Program 2022 - 2023
- Mentor, 2 graduate students, Summer Internship Program 2021, 2022

- Guest lecturer, “Introduction to Statistics and Experimental Design” January 2020
- Co-lead, “Mentoring of Junior Faculty and Fellows” session, Biostatistics and Computational Biology Branch retreat October 2019

University of North Carolina, Chapel Hill, Chapel Hill, NC
- Teaching Assistant, Advanced Epidemiologic Methods (https://github.com/UNC-EPID-722/2016) Spring 2016

Related Professional Experience

Academic and Professional Service

Journal Peer Review
BMJ Open | International Journal of Cancer | Cancer Epidemiology, Biomarkers & Prevention | BMC Public Health | Epidemiology | BMC Pregnancy and Childbirth | Journal of Eating Disorders | Nutrition and Metabolism | Cancer Medicine | Clinical Epigenetics

Society for Epidemiologic Research
- Annual meeting abstract reviewer 2021 - 2022
- Symposium co-chair: “Polygenic Risk Scores: challenges and opportunities for population health equity” December 2020

National Institute of Environmental Health Sciences, Research Triangle Park, NC
- Special volunteer: Publish manuscripts based on research completed during my post-doctoral fellowship. April 2024 – April 2026
- Internal manuscript peer reviewer 2018 - 2022

Professional Development Training
National Institute of Environmental Health Sciences, Research Triangle Park, NC
- Research Mentor Training, 4-week course May 2019, May 2021
- ‘Write Winning NIH Grant Proposals’ 1-day workshop February 2019

Professional Associations
Society for Epidemiologic Research | American Association for Cancer Research